

Method Selection and Planning

ENG1 Cohort 1 Team 3

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Engineering Methodology

We chose to use an agile software engineering methodology as this allowed us the most flexibility as the project is completed and allows for individuals to dynamically contribute to the progress of the system. The advantages of using agile are quick deployment, emphasis on collaboration and the ability to adapt to changing requirements. Quick deployment is important due to the short time scale of the project, collaboration is important to ensure that skills from all members of the team are being utilised to maximum effect and the ability to adapting as requirements change is useful as the features that we choose to implement may change as the project unfolds. The disadvantages of using agile are a smaller emphasis on testing, smaller emphasis on documentation and risk of burn-out. The smaller emphasis on testing can be mitigated by focusing on code quality and adhering to style guides and best practices.

Burn-out can be reduced by allocating work based on team strengths, varying tasks that members do and work in pairs on sprints. These factors make agile the most suitable methodology for our project.

Source Control

GitHub - We chose github for version control as it allows us to remotely collaborate with each other. It allows individual team members to pull code from a centralised repository to work on a section of code before pushing it to a side branch to be reviewed and merged with the main branch of code. This allows for developers to work on code without affecting the work of the other team members and code can be verified before committing it to the main branch, saving time by minimising errors in code.

We also use GitHub projects to keep track of progress throughout the development stage. This allows us to break down large tasks into issues, which can be assigned to individual team members and integrated with pull requests, which aligns with our agile development methodology. We are using Google Drive for some of our documentation and many of our team members are already familiar with GitHub, so we used it for source control.

Development Environment

We chose to use VSCode and IntelliJ in the implementation of our project. The reason why we used these IDEs to program the game is because our team members were most accustomed to these IDEs and they happen to offer reasonable support for the Java programming language. As LibGDX projects generally use their own build scripts, the project itself is mostly agnostic to the IDE used, meaning group members can use their personal preference.

Team Communication

When deciding between software for team communication, the two popular options among our group were Whatsapp and Discord. We chose to use Whatsapp for this project as it is software every member of the group is familiar with.

Frameworks

LibGDX - We chose libGDX as the framework for our game development. It comes with ample features for developing a 2D game. It is widely used and comes with extensive documentation, which means we can get started with a simple setup very quickly. Additionally, LibGDX is Apache 2 licensed, a highly permissive licence that allows the use of their code for any projects with no fees.

Team Organisation

Project Lead - Vadim

- Helping with each of the requirements from the product brief
- Overseeing that team members are completing their responsibilities
- Manage TODO spreadsheet in Google Drive
- Code and documentation quality control
- Task allocation

Documentation - Laina

- Ensuring code documentation is complete
- Create developer README so that developers understand the source code
- Project documentation

Source Control - James

- Maintaining the github repo (branching and merging onto main branch)
- Code reviews (logic errors, implementation of requirements, automated tests, style guidelines, security flaws)

Project structure - Mansi

- Contribute to weekly plans regarding implementation tasks
- Make sure that tasks within the project are not missed

Writeup - Almira

- Making sure the write up is complete
- Summarise team decisions to add to write up
- Proofreading
- Ensure that any methods/programs used have been added to the write up

Quality Assurance - Harry

- Ensure that code is high quality and readable
- Debugging code
- Ensure that game is fun and engaging
- Document known issues with code

We chose to organise the team in this way to allow for clear definition of responsibilities with only small overlaps to avoid confusion as to who is responsible for certain tasks. This allows us to work more efficiently, by reducing time spent on delegating tasks.

Roles were chosen based on our personal interests and skills to maximise participation and mitigate the risk of burnout.

This approach is suitable for developing a small 2D game due to the relatively small scope of each part of the project, meaning one team member is able to lead each main area.

Plan for Key Tasks and Deadlines

Note: Task - priority (Assignee)

#1 Update Risk Assessment - Medium (Laina)

Updating the risk writeup will allow us to include our personal risks of failure that could happen throughout the project so that we ensure our success

#2 Update User Requirements - High (Almira & Laina)

Checking whether we need to update the Requirements write up allows us to add requirements uniquely identified by our team that may have not risen in other teams. As every demand is described in detail and assigned a priority, it allows us to easily plan and assign requirements to different teammates based on their strengths and preferences.

#3 Update the previous Architecture diagram - High (Almira)

Depends on: Score Calculation Implementation (#4), Events Implementation (#5)

This will help us start work on implementation as if we don't have an architecture diagram we do not know how each class of our project should interact with each other. Without this, classes written by different members may not interact correctly.

#4 Score Calculation Implementation - High (James & Harry)

Depends on: Method Selection & Planning (#6)

The map will be made first as it is fundamental to the game and will contain all other entities that are added to the map.

#5 Events Implementation - High (Mansi)

Depends on: Method Selection & Planning (#6)

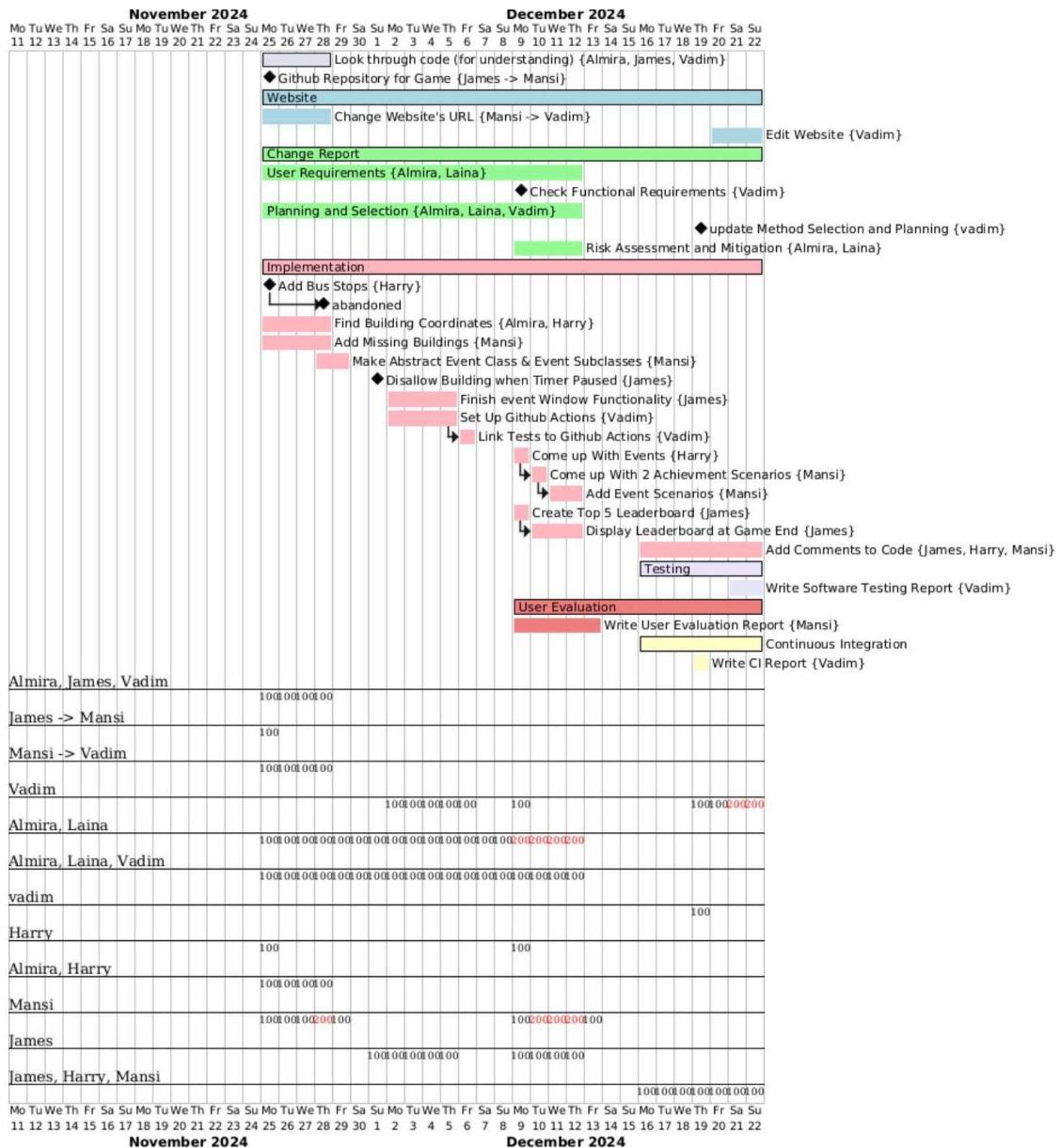
The menu will be made after the map so that it is able to call for the map to be rendered once the play button is pressed

#6 Method Selection & Planning - (Vadim)

The initial version of Method Selection and Planning deliverable must be completed first to give the team a consistent project structure along with allocated roles for them to take. This will avoid confusion during implementation and support separation of labor which will make us more efficient.

WEEKLY PLANS

How the plan changed throughout the project



For the 1st week, we have decided role allocation for the second part of the assessment. We have added one more member of the team who is going to be responsible for implementing the game (Mansi), as we realised that realistically we will need one more person in that part of the project to meet the deadline. Moreover, in the first part of the assessment we did not really have clear role distribution and that is the problem we first tackled.

In the 2nd week, the implementation team (James, Mansi, Harry) started tackling the concerns of implementing score calculation and the way events will be handled in the game. The second part of the team started updating the Team 9's deliverables (Almira, Vadim, Laina) to make sure that the required documents that are written up before implementation is started to give clear project structure to the second part of the assessment. We also have started preparing for user evaluation and setting up continuous integration within our project using Github Actions.

In the 3rd week, we were getting near the Christmas break, so our plan for that week was based around the fact that the team needs to make the most out of the time that we still have together at the campus before the Christmas break. With that idea in mind, James, Harry, and Mansi have put a deadline on the final prototype to be released by Thursday that week, which would allow us later in that day's practical to conduct user testing using that same exact prototype. This week, Almira and Laina have also finished Risk Assessment and User Requirements.

Over the Christmas break, Mansi was supposed to get the User evaluation report done, which was accomplished. No other work was assigned to the other members but progress was made on the Continuous Integration report, Website and Software Testing by Vadim.

After the Christmas break, our plan was to finish any deliverables that were incomplete. According to the plan the remaining work was supposed to be assigned to people to meet a set deadline on Friday, and to ensure equitable work allocation as the workload was realized to be somewhat unbalanced. Taking that into consideration, Harry is meant to work on Software Testing and James on Website and Continuous Integration. Additionally, Almira is finishing the Architecture report. Most of our deliverables are nearing completion. It is highly unlikely that the plan will undergo changes at this stage of the project,